NGUYEN THANH PHU

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SUMMARY

Embedded engineer passionate about low-level programming and hardware-software integration.

EDUCATION

Ho Chi Minh City University of Technology and Engineering — Expected Graduation 2026
GPA: 3.28/4 — Major: Computer Engineering Technology

WORK EXPERIENCE

Ban Vien Corporation, Ho Chi Minh City, Viet Nam — May, 2026 – Present
Embedded Software Engineer

- Maintained Renesas RA-FSP drivers with a primary focus on the RA8 series (Cortex-M85) architecture, including developing and maintaining test suites for DCache-related tickets (pending merge for upcoming releases).
- Resolved customer-reported issues in e² studio, debugging and updating Jinja templates to ensure accurate code generation.

Bosch Vietnam Co., Ltd., Ho Chi Minh City, Viet Nam — Oct, 2025 – Apr, 2026
System Test Engineer Intern

- Developed automated CAPL scripts and a reusable testing library to verify CAN message routing against DBC/ARXML specifications, ensuring queue stability (FIFO/MPFO) under high-load stress testing.
- Cut test execution time by 15% by switching to a batch-checking method, and successfully shared this framework with other projects to speed up setup time.

PROJECTS EXPERIENCE

Zonal ECU Implementation over Ethernet Backbone, Independent Project — 2026

- Architected a modular Zonal ECU on ESP32 (FreeRTOS) with a W5500 Ethernet controller over 20 MHz SPI DMA, implementing a SOME/IP-style UDP communication stack for service discovery, handshake authentication, and telemetry reporting to a PySide6 dashboard.
- Engineered a deterministic 612-byte unified PDU with dual E2E protection (16-bit Checksum + CRC-32) to eliminate IP fragmentation, achieving an average 54.78 ms command-response latency across the Ethernet backbone.
- Eliminated CPU polling starvation by designing an interrupt-driven FreeRTOS pipeline, synchronizing hardware GPIO ISRs, SPI packet dispatch, and priority queues via lightweight Task Notifications and binary semaphores..

xClipboardCapture (Clipboard Manager) on Void Linux, Independent Project — 2026

- Developed a lightweight clipboard manager in C on Linux to automatically save copied text and images, integrated with Rofi for quick history search and retrieval.
- Kept memory usage under 5 MB, allowing the tool to run smoothly in the background without slowing down the system.

TECHNICAL SKILLS

- Programming Languages: C, C++, Python, Bash, CAPL Script.
- Working Environments: Linux (Void Linux, Ubuntu)
- Tools & Platforms: Vector CANoe, Git, GitHub, Qt, Valgrind, Cadence Virtuoso, Altium Designer.
- Protocols: CAN/CAN FD, LIN, UART, SPI, I2C.
- MCU & HW: MCU (ESP32, AT89C52), RPi, FreeRTOS, digital logic design, OOP, DSA.
- Others: Fundamental networking knowledge (NAT, PAT, routing).

CERTIFICATIONS & AWARDS

- Engineering Projects in Community Service (Season 7).
- Samsung Innovation Campus - Internet of Things course.